Noether's Theorem

Every continuous symmetry of the Lagrangian gives rise to a conserved current $j^\mu(x)$ such that

$$\partial_\mu j^\mu = 0 \quad \Longrightarrow \quad \frac{1}{c}\frac{\partial j^0}{\partial t} + \vec{\nabla}\cdot\vec{j} = 0$$

Define charge of a volume V:

$$Q_V = \int_V d^3x \, j^0$$

$$\frac{dQ_V}{dt} = -\int_V d^3x \, \vec{\nabla}\cdot\vec{j} = -\int_{\text{Area}} \vec{j}\cdot d\vec{S}$$

local conservation of charge

To prove the theorem, analyze $\delta\mathcal{L}$ under the symmetry transformation $\delta\varphi_a(x) = X_a(\varphi)$

$$\delta S = \int d^4x \left\{ \frac{\partial \mathcal{L}}{\partial \varphi_a} \delta\varphi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \overbrace{\delta(\partial_\mu \varphi_a)}^{\partial_\mu(\delta\varphi_a)} \right\} =$$

$$= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \varphi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \right] \delta\varphi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \delta\varphi_a \right)$$

We say that transformation $\delta\varphi_a(x) = X_a(\varphi)$ is a <u>symmetry</u> if the <u>Lagrangian</u> $\mathcal{L}$ changes by total derivative $\delta\mathcal{L} = \partial_\mu Z^\mu$ for some $\underline{Z^\mu(\varphi)}$

From above, $\delta\mathcal{L} = \underbrace{\left(\frac{\partial \mathcal{L}}{\partial \varphi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \right)}_{\text{E-L eqn} \Rightarrow \text{satisfied by physical } \varphi \Rightarrow \text{gives zero}} \delta\varphi_a$

$$+ \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \delta\varphi_a \right)$$

$$\Rightarrow \begin{cases} \delta\mathcal{L} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} \delta\varphi_a \right) \\ \delta\mathcal{L} = \partial_\mu Z^\mu \end{cases}$$

symmetry

$$\boxed{\delta\varphi_a = X_a(\varphi)} \Rightarrow \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} X_a(\varphi) - Z^\mu \right) = 0 \Rightarrow \partial_\mu j^\mu = 0$$

transformation

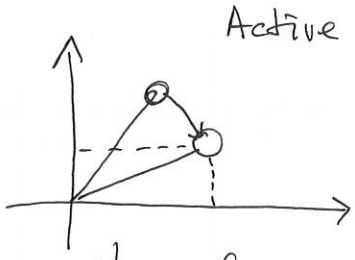
$\Rightarrow$ conserved current $j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_a)} X_a(\varphi) - Z^\mu(\varphi)$

associated with $\underline{\delta\varphi_a = X_a(\varphi)}$

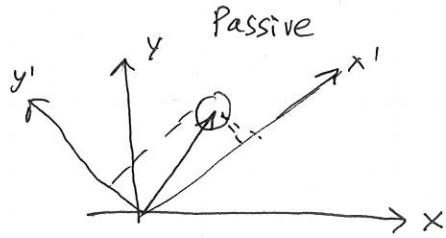
What about space-time transformations?

$\Rightarrow$ energy-momentum tensor $T^{\mu\nu}$

Under transformation $x \to \Lambda x$
fields transform as $\varphi(x) \to \tilde{\varphi}(x) = \varphi(\Lambda^{-1} x)$ } <u>active transformation</u>



physical system is transformed



system of coordinates is transformed

Transformed position
$x' = \Lambda x$
Transformed field
$\phi'(x') = \phi(x)$

<u>Active transformation</u>

suppose x_0 is point of maximum of field in system S.

Then in the tranformed system S'
$x_0' = \Lambda x_0$ is the image of that point.

$\Rightarrow$ transformed $\phi'(x')$ must also have maximum at x_0' !

$$\Rightarrow \underbrace{\phi'(x')}_{\text{in } S'} = \underbrace{\phi(x)}_{\text{in } S}$$

Need to express both sides in S': $\underline{x = \Lambda^{-1} x'}$

$$\Rightarrow \phi'(x') = \phi(\Lambda^{-1} x')$$

or, $\phi'(x) = \phi(\Lambda^{-1} x)$ (dropping "primes").

$\phi(x) \to \phi(\Lambda^{-1} x)$

<u>Passive transformation</u>: $x' = \Lambda x$ $\quad \phi'(x') = \phi(\Lambda x)$

or, $\phi(x) \to \phi(\Lambda x)$

Space-time symmetries : translations

Consider infinitesimal translation $x^\nu \to x'^\nu = x^\nu - \varepsilon^\nu$
where $\nu = 0,1,2,3$ takes ~~some~~ particular value
e.g. $\nu = \alpha$.

$$\phi(x) \to \phi'(x') = \phi(\Lambda^{-1}x) = \phi(x^\nu + \varepsilon^\nu) = \phi(x) + \varepsilon^\nu \frac{\partial \phi}{\partial x^\nu}$$

Lagrangian density changes as well : $\mathcal{L}(x) \to \mathcal{L}(x) + \varepsilon^\nu \partial_\nu \mathcal{L}$
$= \mathcal{L}(x) + \partial_\nu Z^\nu$ where $Z^\nu = \varepsilon^\nu \mathcal{L}$.

<u>Noether's th says</u>

$$\partial_\mu \underbrace{\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \varepsilon^\nu \partial_\nu \phi - \varepsilon^\mu \mathcal{L} \right)}_{J^\mu} = 0$$

$$J^\mu = \varepsilon^\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L} \right)$$

$\varepsilon^\mu = \varepsilon^\nu \delta^\mu_\nu$ ← Kronecker delta, selects $\mu = \nu$.

Want do write $J^\mu = \varepsilon_\nu T^{\mu\nu} = \varepsilon_\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - \boxed{} \mathcal{L} \right)$

what should be here?

$$\boxed{g^{\mu\nu} \delta^\mu_\nu = g^{\mu\nu}}$$

Remember, we raise and low indices with metric $g_{\mu\nu} = g^{\mu\nu} = \mathrm{diag}(1,-1,-1,-1)$

e.g. $\partial^\nu \phi = g^{\nu\mu} \partial_\mu \phi$

$$\Rightarrow T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - g^{\mu\nu} \mathcal{L}$$ Energy-momentum tensor.

Space-time symmetries

Translations and Energy-Momentum tensor

Infinitesimal translation $x^\nu \to x^\nu - \varepsilon^\nu$ $(\nu = 0,1,2,3)$ for some $\underline{\underline{\nu}}$

$$\Rightarrow \phi_a(x) \to \phi_a(\Lambda^{-1}x) = \phi_a(x+\varepsilon) =$$
$$= \phi_a(x) + \varepsilon^\nu \partial_\nu \phi_a(x)$$

* Similarly, the Lagrangian $\mathcal{L}(x) \to \mathcal{L}(x) + \varepsilon^\nu \partial_\nu \mathcal{L}(x) = \mathcal{L} + \partial_\nu F$ changes in the same way (! just as func. of x)

Thus, Noether's Th says that

$$\partial_\mu \Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \underbrace{\delta\phi_a}_{\varepsilon^\nu \partial_\nu \phi_a} - F \Big) = 0 \quad \text{where } F = \varepsilon^\mu \mathcal{L}(x)$$
$$\delta\phi_a = \varepsilon^\mu \partial_\mu \phi_a(x)$$

$$\Rightarrow \text{conserved current } \underbrace{(j^\mu)_\nu}_{T^\mu{}_\nu \,=\, \text{energy-momentum tensor}} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\nu \phi_a - \delta^\mu_\nu \mathcal{L}$$

$$\partial_\mu T^\mu{}_\nu = 0$$

$\boxed{\text{How to go from } \delta^\mu{}_\nu \text{ to } g^{\mu\nu}?}$ mixed tensor

or $$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial^\nu \phi_a - g^{\mu\nu} \mathcal{L}$$

$$\Rightarrow E = \int d^3x \, T^{00} \qquad p^i = \int d^3x \, T^{0i}$$ (are conserved charges, $\underline{\mu=0}$)

let $\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m \phi^2$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi \Rightarrow T^{\mu\nu} = \partial^\mu \phi \, \partial^\nu \phi - g^{\mu\nu} \mathcal{L}$$

$$T^{00} = \partial^0\phi\,\partial^0\phi - \left[\frac{1}{2}\left(\partial_\mu\phi\,\partial^\mu\phi\right) - \frac{1}{2}m\phi^2\right] =$$

$$= (\dot{\phi})^2 - \left[\frac{1}{2}(\dot{\phi})^2 - \frac{1}{2}(\vec{\nabla}\phi)^2 - \frac{1}{2}m\phi^2\right] = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\vec{\nabla}\phi)^2 + \frac{1}{2}m\phi^2$$

$E = \int d^3x\; T^{00}$ indeed.

$$p^i = \int dx\,\partial^0\phi\,\partial^i\phi = \int dx\,\pi\,\partial^i\phi$$

$$\partial^0\phi = \frac{\partial L}{\partial(\partial_0\phi)} = \frac{\partial L}{\partial(\dot{\phi})} = \pi(x) \leftarrow \text{momentum}$$

~~Broken~~ Internal Symmetries

1) $$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi)^2 - \frac{1}{2} m^2 \varphi^2 - \frac{\lambda}{4!}\varphi^4$$

E-L eqn: $$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \right) = 0$$

$$\Rightarrow \partial_\mu \partial^\mu \varphi = -m^2 \varphi + \frac{1}{3!}\lambda \varphi^3$$

But for $\underline{\lambda = 0}$ $$\frac{\partial^2 \varphi}{\partial t^2} - \nabla^2 \varphi + m^2 \varphi = 0$$

$$\varphi(x,t) = \sum_k a_k e^{-i(E_k t - \vec{k}\cdot\vec{x})}$$

$$E_k = \sqrt{k^2 + m^2}$$ Klein-Gordon eq.

2) Two real scalar fields

$$\mathcal{L}_2 = \frac{1}{2}(\partial_\mu \varphi_1)^2 - \frac{1}{2} m^2 \varphi_1^2 + \frac{1}{2}(\partial_\mu \varphi_2)^2 - \frac{1}{2} m^2 \varphi_2^2 - g(\varphi_1^2 + \varphi_2^2)^2$$

Symmetric under "rotation"

$$\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \varphi_1' \\ \varphi_2' \end{pmatrix}$$

this is <u>internal symmetry</u> (operates in $\varphi_{1,2}$ space)

$$\begin{pmatrix} \partial_\mu \varphi_1 \\ \partial_\mu \varphi_2 \end{pmatrix} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix} \begin{pmatrix} \partial_\mu \varphi_1' \\ \partial_\mu \varphi_2' \end{pmatrix}$$

$$\varphi_1^2 + \varphi_2^2 = (c\varphi_1' - s\varphi_2')^2 + (s\varphi_1' + c\varphi_2')^2 = \varphi_1'^2 + \varphi_2'^2$$

$$\Rightarrow \mathcal{L}_2(\varphi_{1,2}) = \mathcal{L}_2(\varphi_{1,2}')$$

Introduce $\begin{cases} \psi = \frac{1}{\sqrt{2}}(\varphi_1 + i\varphi_2) \\ \psi^* = \frac{1}{\sqrt{2}}(\varphi_1 - i\varphi_2) \end{cases}$ $\begin{cases} \varphi_1 = \frac{1}{\sqrt{2}}(\psi + \psi^*) \\ \varphi_2 = \frac{1}{\sqrt{2}i}(\psi - \psi^*) \end{cases}$

$$\varphi_1^2 + \varphi_2^2 = \frac{1}{2}(\psi + \psi^*)^2 + \frac{1}{2i^2}(\psi - \psi^*)^2 =$$

$$= \frac{1}{2}\left(\psi^2 + \psi^{*2} + 2\psi\psi^* - (\psi^2 + \psi^{*2} - 2\psi\psi^*)\right)$$

$$= 2|\psi|^2 = 2\psi^*\psi$$

$$(\partial_\mu \varphi_1)^2 + (\partial_\mu \varphi_2)^2 = \frac{1}{2}(\partial_\mu \psi + \partial_\mu \psi^*)^2 - \frac{1}{2}(\partial_\mu \psi - \partial_\mu \psi^*)^2$$

$$= \frac{1}{2} \cdot 4\, \partial_\mu \psi^* \partial_\mu \psi$$

$$\Rightarrow \mathcal{L}_2(\varphi_1, \varphi_2) = \partial_\mu \psi^* \partial^\mu \psi - m^2|\psi|^2 - 4g|\psi|^4 = \mathcal{L}(\psi)$$

The potential $V(\psi) = m^2|\psi|^2 + 4g|\psi|^4 = V(|\psi|)$
is invariant under $\begin{cases} \psi \to e^{i\alpha}\psi \\ \psi^* \to e^{-i\alpha}\psi^* \end{cases}$ $|\psi|^2 \to |\psi|^2$
$U(1)$ symmetry

Check now $\partial_\mu \psi^* \partial^\mu \psi \to \partial_\mu(e^{-i\alpha}\psi^*)\, \partial^\mu(e^{i\alpha}\psi) =$

$= \partial_\mu \psi^* \partial^\mu \psi$ <u>for $\alpha = \text{const}$</u>

$\Rightarrow$ $\mathcal{L}(\psi) = \mathcal{L}(\psi e^{i\alpha})$ is invariant under the transformation

$\Rightarrow$ $\underline{\delta\mathcal{L} = 0}$

Infinitesimal form of transformation ($\alpha \to 0$)

$$\begin{cases} \psi = e^{i\alpha}\psi' = (1+i\alpha)\tilde{\psi} \\ \psi^* = (1-i\alpha)\tilde{\psi}^* \end{cases} \Rightarrow \delta\psi = i\alpha\psi$$

in more details

$$\left[\begin{array}{l} \psi - \tilde{\psi} = i\alpha\tilde{\psi} \\ \delta\psi = i\alpha\psi \end{array}\right.$$

because to linear order in α $\psi = \tilde{\psi}$]

$$\Rightarrow \frac{1}{\sqrt{2}}(\psi + \psi^*) = \frac{1}{\sqrt{2}}\left[(1+i\alpha)\tilde{\psi} + (1-i\alpha)\tilde{\psi}^*\right] =$$

$$= \frac{1}{\sqrt{2}}\left[\tilde{\psi} + \tilde{\psi}^* + i\alpha(\tilde{\psi} - \tilde{\psi}^*)\right]$$

$$\Rightarrow \varphi_1 \to \tilde{\varphi}_1 - \alpha\tilde{\varphi}_2$$

$$\frac{1}{\sqrt{2}i}(\psi - \psi^*) = \frac{1}{\sqrt{2}i}\left(\tilde{\psi} - \tilde{\psi}^* + i\alpha(\tilde{\psi} + \tilde{\psi}^*)\right)$$

$$\varphi_2 \to \tilde{\varphi}_2 + \alpha\tilde{\varphi}_1$$

$$\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} = \begin{pmatrix} 1 & -\alpha \\ \alpha & 1 \end{pmatrix}\begin{pmatrix} \tilde{\varphi}_1 \\ \tilde{\varphi}_2 \end{pmatrix}$$

Compare with $\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} = \begin{pmatrix} 1 & -\theta \\ \theta & 1 \end{pmatrix}\begin{pmatrix} \tilde{\varphi}_1 \\ \tilde{\varphi}_2 \end{pmatrix}$ for $\theta \to 0$

$\Rightarrow \psi = e^{i\alpha}\tilde{\psi}$ is equivalent to $\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix}$ rotation.

$$\delta\mathcal{L} = 0$$

What is the corresponding current?

$$j^\mu \sim \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} i\alpha \psi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi^*)} (-i\alpha \psi^*)$$

Since $\alpha \to 0$ is arbitrary $\to$ remove it

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} = \partial^\mu \psi^* \quad , \quad \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi^*)} = \partial^\mu \psi$$

$$\Rightarrow j^\mu = i(\partial^\mu \psi^*)\psi - i\psi^*(\partial^\mu \psi)$$ conserved current

Conserved charge $Q = \int d^3x\, j^0 = i\int d^3x \left((\partial^0 \psi^*)\psi - (\partial^0 \psi)\psi^*\right)$

<u>Non-relativistic field theory</u>

$$\mathcal{L} = i\psi^* \partial_0 \psi - \frac{1}{2m} \nabla\psi^* \nabla\psi - 4g|\psi|^4 - m|\psi|^2$$

$$\Rightarrow j^0 = \frac{\partial L}{\partial(\partial_0 \psi)}(i\psi) + \frac{\partial L}{\partial(\partial_0 \psi^*)}(-i\psi^*) = i\psi^*(i\psi) + (0)(-i\psi^*) = -|\psi|^2$$

$Q = \int d^3x\, |\psi|^2$ is conserved $\leftarrow$ TOTAL charge!